MD NASIF RAFIDI

AI and NLP Research | Medical Imaging | Python and SQL

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Profile

CSE undergraduate at American International University-Bangladesh with published IEEE research on multilingual generative NLP and contributions to deep learning for medical imaging. Research portfolio includes Bangla text analysis, human activity recognition and time-series forecasting. Best Thesis Defense Poster Award recipient with programming, database and technical documentation experience. Research interests include LLM efficiency, model evaluation and healthcare AI.

Education

American International University-Bangladesh

BSc in Computer Science and Engineering | 2023-Present | CGPA 3.41

Academic focus: Data Science, Data Mining and Warehouse, Advanced DBMS, Computer Vision and Pattern Recognition, and Natural Language Processing.

Published Research

Cache-Aware Generative NLP: Reducing Latency in Large-Scale Multilingual Systems

IEEE QPAIN 2026

Co-author | Published in IEEE Xplore

DOI 10.1109/QPAIN69676.2026.11546324

Study of Adaptive Linguistic Computation Caching for multilingual generation; the research reports 75.8% latency reduction across 12 evaluated models.

Accepted Conference Paper

Attention-Enhanced CNN-GRU Framework for Short-Term Grid-Tied Solar Power Forecasting Using Real Meteorological Time-Series Data

International Conference on Data Science, AI and Applications (ICDSAIA 2026)

Accepted for presentation | Paper 161 | Acceptance notice 24 September 2026

Attention-based CNN-GRU research for forecasting solar power from meteorological time-series data.

Conference Manuscripts with Revision Decisions

A Decision-Support Framework for Bangla Civic-Complaint Triage: Category-Conditioned Routing and Calibrated Priority Ranking

International Conference on Decision Aid Sciences and Applications (DASA 2026)

Accepted with major revisions | Paper 1571356484 | Decision 11 October 2026

Bangla NLP research on complaint categorization, routing and priority ranking.

A Leakage-Aware Comparative Analysis of Machine Learning Models for IMU-Based Human Activity Recognition Systems

International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS 2026)

Conditionally accepted | Paper 193 | Decision 25 September 2026

Comparative study of machine learning methods and data leakage in sensor-based activity recognition.

Further Research and Conference Activity

CAMTS-UNet | LGG MRI Segmentation

- Contributed to deep learning research on low-grade glioma MRI segmentation using attention and multi-scale U-Net concepts.

Climate-Adaptive Energy Prediction for Electric Vehicles in Bangladesh: An LSTM Approach

- Co-authored research on LSTM-based energy prediction; presented by the research team at ICEI 2026.

IEEE BECITHCON 2026

- Co-author of accepted papers 223 and 239 in the official technical programme; listed as the scheduled presenter for paper 223 in Health Informatics.

IEEE i-COSTE 2026

- Accepted conference research contribution at the International Conference on Sustainable Technology and Engineering.

Technical Skills

Programming and databases: Python, Java, C++, SQL, Oracle SQL, PL/SQL and MySQL.

Machine learning framework: PyTorch.

Data and research: Data cleaning and preparation, data analysis, data mining, NLP, computer vision, literature review and technical documentation.

Development: HTML, Tailwind CSS, JavaScript, PHP, application debugging and database design.

Tools: GitHub, LaTeX, Advanced Excel, Microsoft Office and Canva.

Professional Experience

IT Manager and Web Developer | Epitaxial Marketing | Oct 2024-2025

- Managed databases, records and reports, maintained data accuracy, and supported daily digital operations.

App Developer Intern | Programme - Unreal Engine | Oct 2024-2025

- Built and tested application modules, debugged system issues, and followed structured development workflows.

IT Support and Web Volunteer | IEEE AIUB Student Branch | Current

- Supported web and event technology operations, troubleshooting and technical readiness.

Selected Software Projects

University Payroll Management System | Oracle SQL and PL/SQL

- Designed a 3NF database and implemented procedures, functions, cursors, triggers, packages, locking, exception handling and transaction management.

University and Library Management Systems | Java and MySQL

- Developed systems for structured records, database operations and management workflows.

Awards and Training

- Best Thesis Defense Poster Award | American International University-Bangladesh | AI/ML in Finance | 6 August 2026.
- First Runner-Up, Quest for Science | Bangladesh Science Exposium 2025 | AIUB team member.

Training: Data Solution 360 - Advanced Excel, data handling, automation and reporting; Aspire Leadership Program, Aspire Institute.

Languages: Bangla (native); English, Hindi and Arabic (intermediate).